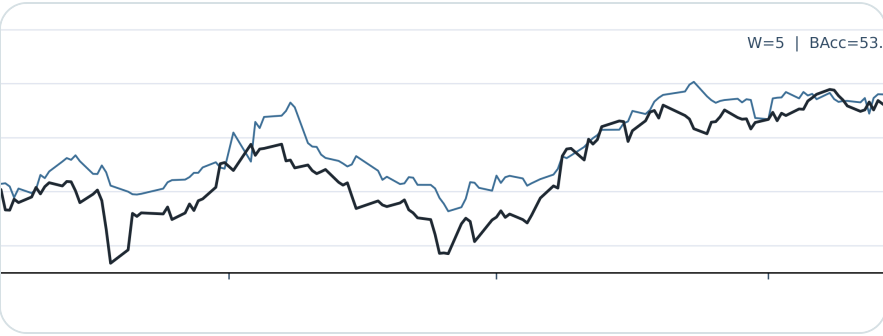


1 INPUTS

POINT-IN-TIME MARKET DATA

SET50 daily index

2012-2025 116 TA features t → t+1



Observed and forecast traces are evaluated only after the next session becomes available.



SECOND MODALITY Dated financial news

Out-of-sample sentiment with date, ticker and relevance controls

2 POINT-IN-TIME RELIABILITY AUDIT

Expanding-window outer evaluation

Train-only transforms; purged labels



Five frozen architectures

LSTM CNN LSTM-CNN LSTM-Attention LSTM-CNN-Attention



Causal VMD

Rolling 60-day decomposition vs matched Full-TA inputs



News and LLM

Market-only controls plus Bull/Bear/Leader sentiment audit



Regime and XAI

Bull/Sideway/Bear routing; train-only SHAP and LIME fidelity check



Forward and transfer

Later-period stress test and frozen SET100 robustness check

Next-session direction UP or DOWN

3 KEY EVIDENCE

53.64%

Highest mean balanced accuracy

LSTM-CNN-Attention across four held-out years and five seeds

+1.46 pp

CNN Regime-SHAP

Selective forecasting gain; not shared by all architectures

+5.93/+6.00 pp

Leader sentiment

Versus equal-call and near-cost controls; intrinsic endpoint

Robustness findings

-0.60 to +0.35 pp VMD effect across architectures

- Predicted news and SET100 transfer showed no consistent directional gain.
- Benefits depended on architecture, endpoint and evaluation period.

Reliable conclusions require temporal, falsification and transfer checks.